The Research Facility on the Moon

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ABSTRACT

The research facility on the Moon is an experimental, structural design for space exploration. It was developed as the Student Master Thesis at Cracow University of Technology, Faculty of Architecture. The main goal was to introduce innovative extraterrestrial architecture. Modularity, multifunctionality, and simplicity of the concept were key objectives. It was crucial to ensure the safety of the lunar crew. The whole range of threats and problems have been solved. The facility is formed from seven modules. The construction of each of them consists of repetitive, easy to transport components. Project is prepared to host twelve astronauts, but it is expected to expand in the future. This structural concept for lunar architecture is universal and could be useful in other, different circumstances.

INTRODUCTION

Historical background. We, humans have been living on planet Earth for hundreds of thousands of years. We looked at the starry sky, concerned with mystical questions. The history of spaceflight lasts nearly eighty years. It was on June 20th in 1944, when German V2 rocket crossed the Karman line and became the first man-made object to travel into space, Rogers (2008). On October 4th in 1957, Sputnik 1 was launched into an elliptical low Earth orbit. It was the first artificial terrestrial satellite. In 1961, on April 12th Russian cosmonaut Jurij Gagarin became first man to orbit the Earth. This mission, Wostok 1, was first manned flight into space, Bond (2003). Almost a year later, in 1962, on February 20th, astronaut John Glenn became the first American in the orbit, NASA (2006). Finally in July 1969 man landed on the Moon. It was first out of few crewed missions on the lunar surface, NASA (2019). The last one was in December 1972. Since then, robots and probes have taken care of the Moon exploration. Recently, topic of lunar experiments and extraterrestrial design became popular again. Research on the Moon facilities, transport possibilities, 3d-printing technology, new materials and space ergonomy has taken a faster turn. Architecture and studies in this field can freely go beyond the borders of reality and break away from planet's surface.

Concept. The idea of described here, crewed base on the Moon was born in the Cracow University of Technology, in February 2019, and became topic of student master thesis. The work was done at the Department of Construction and Building Techniques, Faculty of Architecture, Cracow University of Technology. It was a great challenge and fascinating adventure into the stars. To begin with, difficulties related to transport defined strict requirements regarding the mass, dimensions and form of the concept. The Moon environment is hostile and unfriendly. Gravity is six times smaller than on the Earth, there is no atmosphere or regular electromagnetic field. Meteors pulled by gravity, do not encounter friction that could slow them down before hitting the ground. Furthermore, solar radiation can not be filtered at all, therefore lethal rays reach surface of the Moon. What's more, the large temperature amplitude

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between lit and dark areas is a challenge for the designer. These are significant threats to potential lunar building. To begin with, examination of existing concepts and ideas had to be done. It was crucial to understand already accepted and acknowledged solutions. Among most famous futuristic, extraterrestrial projects in the field of architecture, are designs of the Foster+Partners for European Space Agency. In 2012 the practice took part in research on building possibilities from lunar soil, known as regolith and introduced concept of the moon base, Foster + Partners (2012). Object is capable of housing four people, and offers protection from deadly radiation, temperatures and meteor showers. Next in 2015, practice continued their work and took part in competition for a 3d-printed modular habitat on Mars, Foster + Partners (2015). The research on others work resulted in some assumptions. Significantly lower gravity of the Moon was judged as sufficient for temporary stay. Size of lunar crew for basic facility (first stage of colonization) was estimated at 10 to 15 astronauts. Entire project had three major development directions. The first aim was to find proper, innovative structural solution. Modularity, ease of transport, simplicity and flexibility became most valuable features of searched form. The second aim was to protect the structure from potential external damage. The third aim was to provide functional proposal for final concept. When all major problems and objectives had been pointed out and described, seeking out for solutions started. Whole variety of methods had been involved.

METHODS

Hand drawing. The first method used was focused on visual stimuli. To avoid cage of technical limitations in designing process, hand drawing had been used. Sketches are the fastest and simplest method. It is the best way of visualising experimental ideas and ephemeral thoughts. Series of sketches, diagrams, illustrations and pictures were born in the process. When every notion can get so easily manually poured onto paper, designing becomes truly free. Potential of this method is limited only by imagination. The most significant conclusion was defining spherical shape as most efficient. First conceptual design had been created (see Figure 1).

Virtual model. The second method was focused on geometrical aspect. Virtual models of first basic ideas were created with help of computer. This way of designing was naturally next step after hand drawing. It was crucial to put two dimensional assumptions into three dimensional virtual environment. Visualising the geometry took place in Blender 3D, open-source graphic computer program. It gave whole variety of new possibilities. Precise control of shape, scale, dimensions and modularity of concept made the real breakthrough in design process. Both second and third approach were key factors to establish first forms and systems. The most significant conclusion was defining platonic solids as most useful. Matter of modularity was introduced as truncated icosahedrons connected with each other in structural systems resembling crystal patterns (see Figure 2).

3d printing. The third method was destined to verify results of virtual design. The idea was to put artificial, virtual geometry into physical world. Three dimensional computer generated model was printed out in certain scale (see Figure 3). It was clearly most obvious way to judge concept output. Possibility of 3d printing and experiments has been provided by Department of Descriptive Geometry and Digital Technologies, Faculty of Architecture, Cracow University of Technology.

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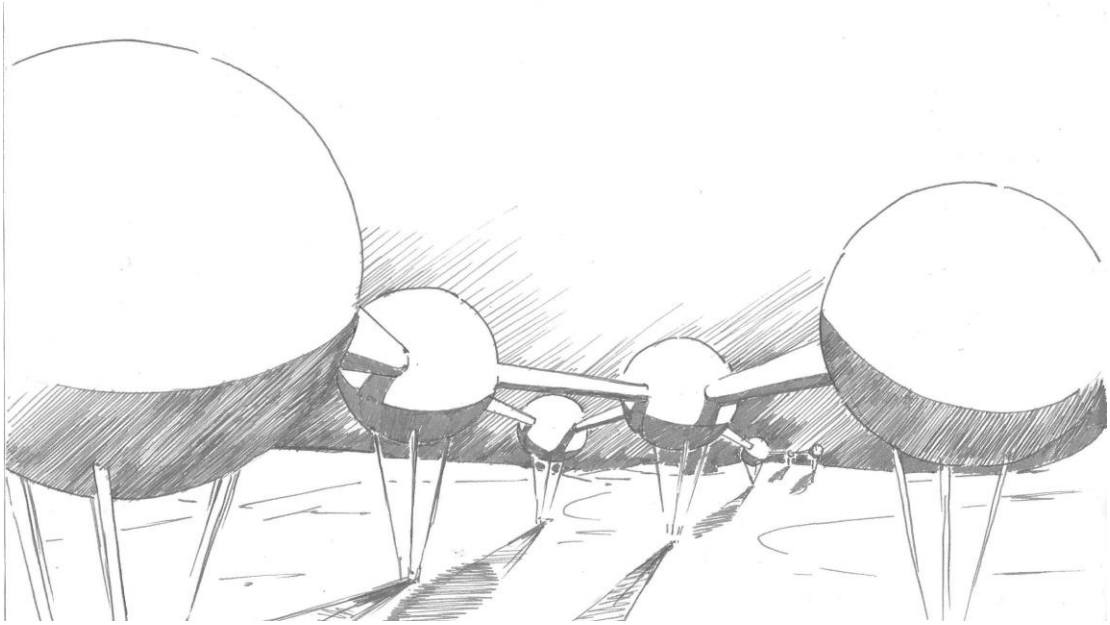

Figure 1. Early design sketch.

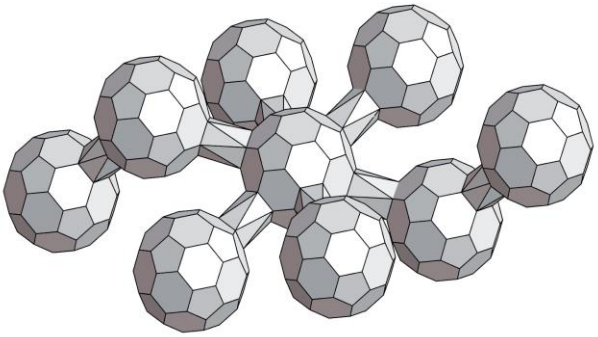

Figure 2. First structure consisting of truncated icosahedrons.

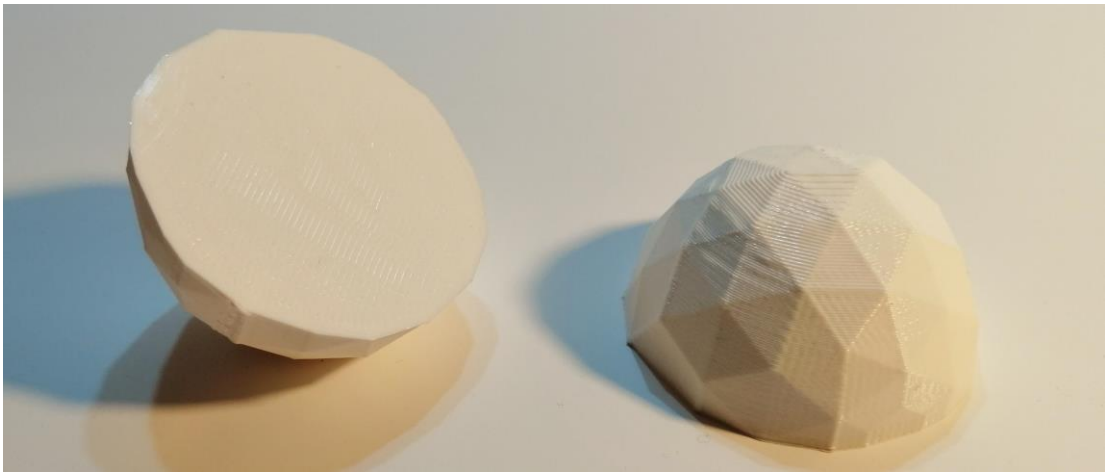

Figure 3. 3d printed components of physical model.

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Technical drawings. Final method of design involved technical drawings. With usage of virtual models and CAD programs, flat schemes were created. Aim was to produce detailed sections and floorplans of project. This approach ensures clarity and readability of concept. This way results at any stage could be understandable and easy to correct. Two dimensions simplified complex solids. Measurable sections and floor plans helped determine functional interior.

RESULTS

As far as the whole research was divided into three major aspects, three key results were expected- construction, protection and function.

Construction. First outcome was entirely focused on structure. There were two main directions in analyzing construction. Concept could have been based on inflatable structures, or consisting of repetitive, rigid assembly elements. The main reason for consideration were transport difficulties. Launching elements into space is a major complication. Therefore size and weight of construction were significant. Important disadvantage of inflatable concept was higher risk for crew when losing the tightness. Yet on the other hand, rigid structures are heavier and more complex. Eventually second option has been chosen. Besides behaviour of stiff construction is more predictable in altered gravity extraterrestrial conditions. When it comes down to form, it was clear that spherical and cylindrical shapes have best pressure distribution and good volume to surface ratio. Hand drawings and virtual modelling directed research to truncated icosahedron. Even distribution of twelve regular pentagons and twenty regular hexagons provided expected repeatability and divisibility. Further step was triangulating each face of designed solid (see Figure 4).

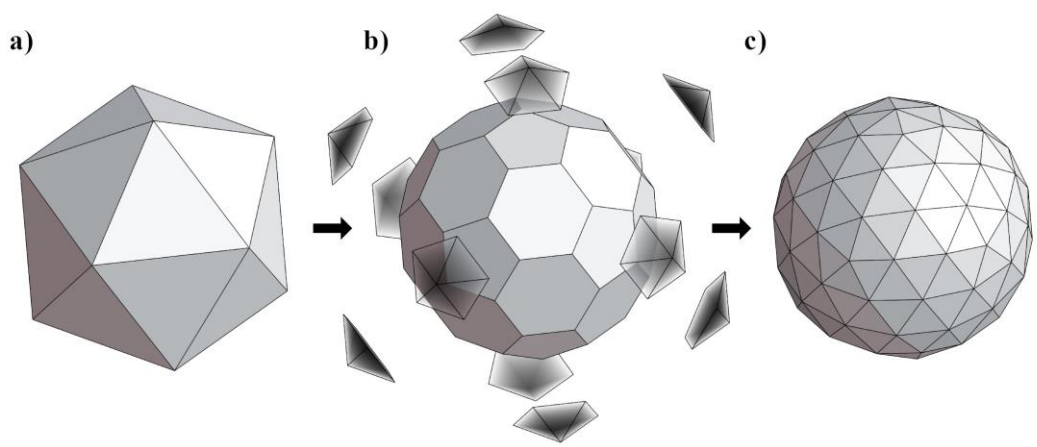

Figure 4. a) Icosahedron, b) Truncated icosahedron, c) Triangulation.

A light rod construction system has been chosen. Established exoskeleton has been reinforced with internal support bars. Six load-bearing rods connected vertices of two opposite hexagonal faces of fromer truncated icosahedron. This spherical construction origins from icosahedron, and because of that, can be divided into twenty repetitive elements (see Figure 5). Line of division is defined by edges of former icosahedron. Every single piece out of this twenty, consists of nine flat triangles, forming curved, triangular shape. Possibility of mounting modular spherical polyhedron from regular parts, actually solved transport issues. Secondly, corridor

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modules have been designed. Main aim was to make use of selected faces of the former truncated icosahedron. There were two options: regular pentagons and regular hexagons. Due to asymmetry of icosahedron, elongated antiprisms appeared to be the best solution (see Figure 6).

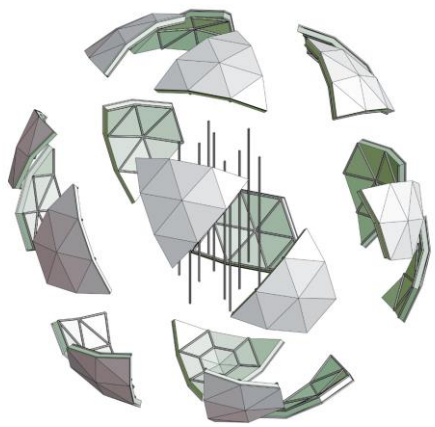

Figure 5. Single module construction explosion.

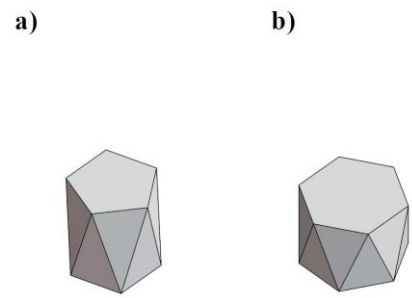

Figure 6. a) Pentagonal corridor, b) Hexagonal corridor.

Edge of designed truncated icosahedron is 2.10 m long. Diameter of each spherical module is 10.02 m. Accessible volume is estimated at 520 m^3. Edge of pentagonal and hexagonal faces of corridor modules are correspondingly 2.1 m long. Meanwhile pentagonal component is 4 m long thus creating volume of 30.35 m^3 and hexagonal component is 2.5 m long thus creating volume of 28.64 m^3. Mentioned dimensions, naturally refer to construction. Unlimited additivity of spherical modules to each other, with corridor modules as connection nodes, was key to the concept. Finally, most important result of computer analysis on virtual model was multiplicity of expansion possibilities. Corridor module can be attached to spherical module at any pentagonal face of the former truncated icosahedron. Of course each sphere requires individual triangulation guidelines, to ensure suitable mounting holes. Basically, due to number of regular pentagons in truncated icosahedron, there are twelve possible locations for pentagonal corridors and twenty for hexagonal corridors. Structures built with those two repetitive components provide flexibility and multifunctionality. Because this spherical construction has its origins in platonic solid, expansion of modules is extremely regular. It resembles system of crystal structures. Eventually whole project can be divided into stages. It can grow in size according to number of occurrences of each component (see Figure 7).

Protection. Second outcome was focused on defining protection system for the whole structure. Project is destined to be located in adverse conditions. Main threats are solar radiation, possible meteor showers and temperature amplitude. Designing multilayer shell over construction appeared to be most effective approach. Main objective was to adapt vacuum as perfect thermal insulator. Detailed solution is based on three layers. Internal thermal insulation is focused on stopping the escape of heat from within the structure. This layer is 20 cm thick and placed on top of rod construction. On the outside of module is external, thin layer destined to absorb temperature differences and deadly solar radiation. Its thickness is 5 cm. In between those two layers, there is narrow space filled with vacuum as the insulator. The distance between them is equal 5 cm. This technology is destined to face environmental conditions in the first phase.

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The next step is to cover finished structure with regolith. Moon dust has radiation-absorbing properties and protects against meteor showers. Finding application for materials available on site is one of the most important research goals for the lunar crew.

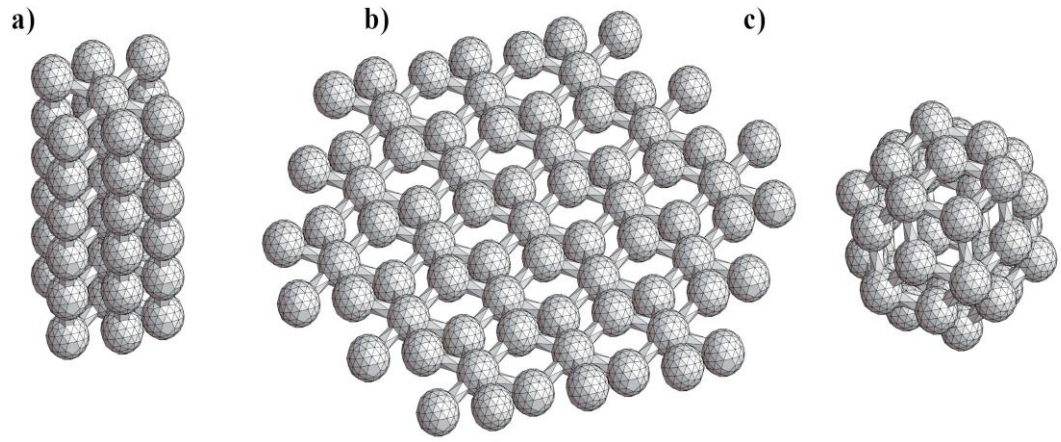

Figure 7. a) Vertical expansion, b) Horizontal expansion, c) Spherical expansion.

Function. Third outcome was focused on creating functional proposal for designed volume. Eventually, three flexible modules have been introduced. They differ only in the number of corridor exits and floors arrangement. Regarding the dimensions of each module, and estimated crew size, basic concept considers seven spherical modules. Six out of those are placed in a circle. Three are placed higher and three are placed lower. Last component is placed centrically, between others, defined as common space and communication node. Whole structure stands on light posts in order to minimize contact with the ground. Modules put on top are designed as living and sleeping area for crew members. Every single one can house four astronauts and consists of common room, bathroom and four individual rooms. Each private area is equipped with with bed, desk, wardrobe and small storage space. Modules put on bottom are more diverse. First one consists of dining area, kitchen and space to store food. The second is essentially a laboratory sector. It covers two floors with scientific equipment for carrying out various experiments. The third one enables crew physical activity. This space gym ensures astronauts will remain in good health. Theoretically, a crew of twelve could use these three sections in turns. Inhabitants of the moon base can either spend time in company or alone. They can organise day cycle in balanced and secure way. Both physical and psychological aspects of the mission have been taken under consideration.

Energy, supplies and materials. The research was mainly focused on structural aspect, therefore it is assumed that already existing technology and solutions meet other needs of the crew. Neither technical drawings or models, show materials or complicated mechanisms in detail. Hatches, surface materials, specific furniture solutions and life support systems are examples of aspects only briefly analyzed. Nevertheless, there is certain space in design predicted for installation of air generator, laboratory equipment, fluid management, ventilation system and all kind of furnishings. So called, technical areas are located in the bottom and top of each spherical module and in the central, construction shafts. Solar energy is expected to be primary power source, whilst food could be regularly sent from Earth.

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The research resulted in set of sketches, technical drawings and models. Early stage of the design process provided various hand-drawn concepts. The project has been presented in technical drawings with varying level of detail to ensure universal readability of the concept (see Figure 8). The virtual model has taken geometric experiments to the new level. Moreover, it facilitated the creation of computer visualizations of the designed object (see Figure 9). Eventually the mock-up of the complex has been 3d printed on a scale of 1:200.

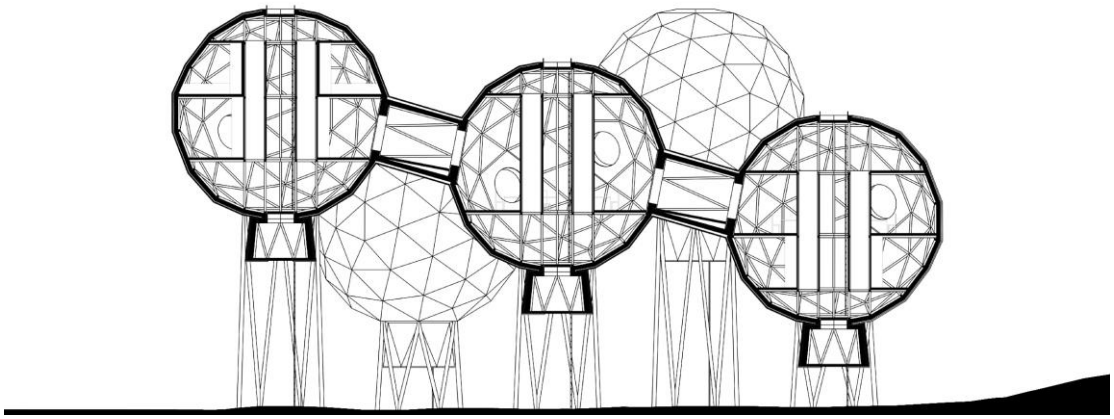

Figure 8. Section of the complex - technical drawing, original scale 1:100.

Figure 9. Facade of the complex- computer visualisation.

DISCUSSION

Final result is an example of innovative, extraterrestrial architecture. The developed concept could be compared to Foster + Partners Lunar Habitation project in order to prove its potential. Due to the three main directions of the described design process, comparison will be made accordingly. From structural point of view, concepts differ a lot. Facility designed by Foster + Partners is packed in tubular transport module. It can be unfolded and inflated. On the organic shaped dome created in this way, the robot prints layers of regolith. Cellular structure of this protective thick shell, resembles foam. Second concept is assembled from repetitive rigid components. Modules are regular polyhedrons. Structure is expected to be covered with regolith

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but not in the very first stage. At the beginning construction would be standing on lean posts to keep buildings above the ground. Yet both concepts respond to the transport difficulties. Secondly, the protection from temperature amplitude, deadly solar radiation and meteor showers is provided. It is worth pointing out that, both structures are multi-layered. It is interesting that the estimated number of astronauts living in each module is the same in both projects. This undeniably confirms the ratio of the size to the maximum transport volume. Each concept assumes expansion of buildings and provides the possibility of additive growth of number of modules.

CONCLUSION

To sum up, the initial design was destined to be placed on the Moon, but main construction is multipotential. The structural concept can be adjusted to various environmental conditions. This project assumes that the main part of the Moon research facility will consist of seven spherical modules, but expansion of this complex is almost unlimited. Furthermore, transport issues have been resolved thanks to modules made of repetitive components. The concept meets physical and structural requirements of lunar environment. Problems of temperature amplitude and solar radiation are solved. Functional design is flexible and adjustable. Whole research proves potential of regular polyhedrons in extraterrestrial architecture and provides innovative proposal – The Research Facility On The Moon (see Figure 10).

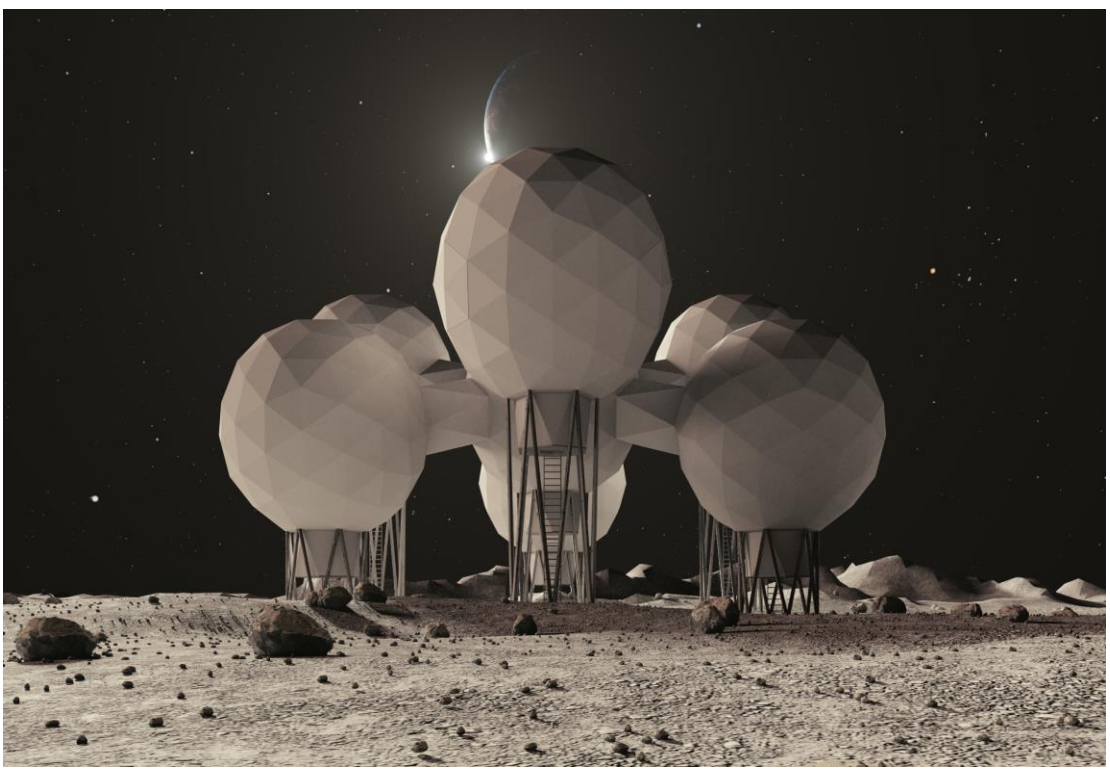

Figure 10. Final project - computer visualisation.

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